

Miaomiao Chen

Sr. Machine Learning Engineer, AstraZeneca PLC. | Ph.D.

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SUMMARY

ML Engineer with 7+ years of experience designing, deploying, and optimizing generative-AI and retrieval-based systems. Skilled in Python, PyTorch, and FastAPI, with strong foundations in data structures, algorithmic optimization, and large-scale backend development. Proven success in building low-latency LLM pipelines, AI agentic frameworks, and production MLOps infrastructure for real-world applications.

Fun fact: National Chess Champion (Multiple Titles, China)

EDUCATION

M.S., Computer Science, Georgia Tech, GA (Online Program)	2024- present
Ph.D., Chemistry, University of Virginia, VA	2024
B.S., Bioscience, ShanghaiTech University, China	2019

SKILLS

Programming: Python, SQL, Spark

Machine Learning: Transformers, LLMs, Reinforcement Learning, Agentic AI, Pytorch, RecSys (RAG, HSTU)

WORK EXPERIENCE

Machine Learning Engineer (formerly Sr. Data Scientist), AstraZeneca Pharmaceuticals PLC., MD July 2024 -present

- Led the end-to-end development of robust **MLOps** pipeline, enabling custom decision metrics and transforming unstructured patient data into actionable insights, resulting in a 60% precision improvement over commercial tools.
- Engineered a custom patient-data embedding (**NMF, Two-Tower**) and fine-tuned an Encoder/Decoder-based **foundation model** via transfer learning, achieving 0.20 uplift in ROC AUC for clinical drug-response prediction.
- Deployed Python Flask/FastAPI applications for real-time clinical data quality monitoring, enabling cross-functional teams to optimize workflows and enhance data pipeline reliability and routine work efficiency by 70%.

PhD researcher, University of Virginia, VA 2019-2024

- Designed multiple projects applying machine learning based algorithms (Random Forest, XGBoost, LSTM) to down select drug candidates with 85.3% accuracy in drug classification, saving months of time in blind testing.
- Conducted large-scale exploratory data analysis and designed A/B tests to evaluate novel anti-inflammation drug effects, significantly contributing to publications in top journals as first author of PNAS (Proceedings of the National Academy of Sciences).

PROJECTS

Krono: Voice-Activated AI News Agent | LLM Systems, GRPO, MLOps [[GitHub](#)][[Demo](#)]

- Built and launched a production voice-based AI news agent using LangGraph-based agent orchestration, enabling conversational access to real-time and recent market news and reducing user information lookup time by ~70%.
- Designed and operated an agentic ML system (intent understanding, tool selection, retrieval, ranking, response generation) optimized for seconds-level end-to-end latency and reliable performance under concurrent usage.
- Post-trained a Qwen-0.5B model with GRPO (verl) for large-scale news classification and routing, cutting LLM API usage by ~60% and inference cost by ~50% while preserving response quality.

Generative Retrieval-based Short Video Recommendation System | HSTU, RoPE [[GitHub](#)]

- Designed and implemented a scalable SOTA transformer-based (HSTU) recommendation algorithm for long-sequence user behavior in video recommendation, improving HR@k and NDCG@k by 11% on KuaiRec 2.0 compared to SASRec.
- Applied an encoder-decoder model with point-wise generation to predict the next user-preferred video embedding.
- Customized RoPE embeddings, boosting NDCG@K by 15% over standard positional encoding.
- Enhanced the base HSTU model by incorporating self-adaptive rating, yielding a 3% increase in HR@k and NDCG@k.

Flash Amazon RecSys | Deep Learning, Two-Tower, DCN/FM [[GitHub](#)]

- Built a two-stage RecSys pipeline (Recall + Ranking) with Two-Tower retrieval, FM, and DCN ranking models; achieved 31x improvement in Recall@5 using pair-wise contrastive training and 62.8% HR@50 end-to-end.
- Designed a hierarchical vector-search engine for 1M+ product embeddings using cluster-based ANN indexing, enabling sub-10ms retrieval latency and scalable candidate generation.
- Implemented full training/eval framework in PyTorch with multi-objective loss, vector-DB construction, and ranking metrics (HR@K, NDCG); delivered production-ready recommendation service with <50ms total inference time.